Q2 Kinematic graphs

Name:

Class:

Date:

Time: **284 minutes**

Marks: **236 marks**

Comments:

Q1.

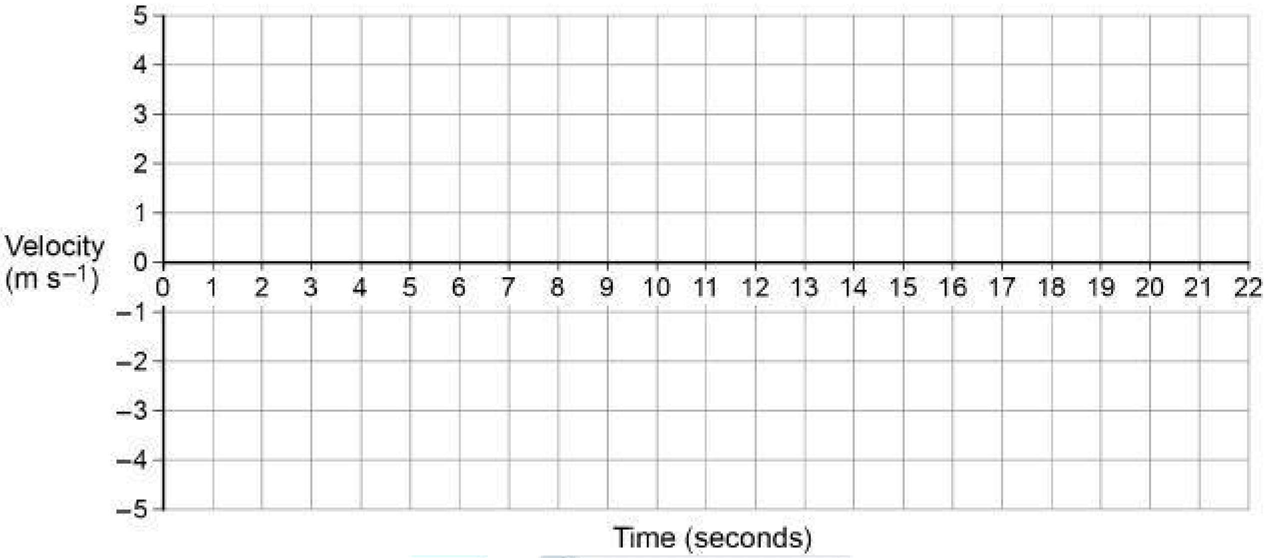
A vehicle, which begins at rest at point *P*, is travelling in a straight line.

For the first 4 seconds the vehicle moves with a constant acceleration of 0.75 m s^{-2}

For the next 5 seconds the vehicle moves with a constant acceleration of −1.2 m s^{-2}

The vehicle then immediately stops accelerating, and travels a further 33 m at constant speed.

(a) Draw a velocity−time graph for this journey on the grid below.



(3)

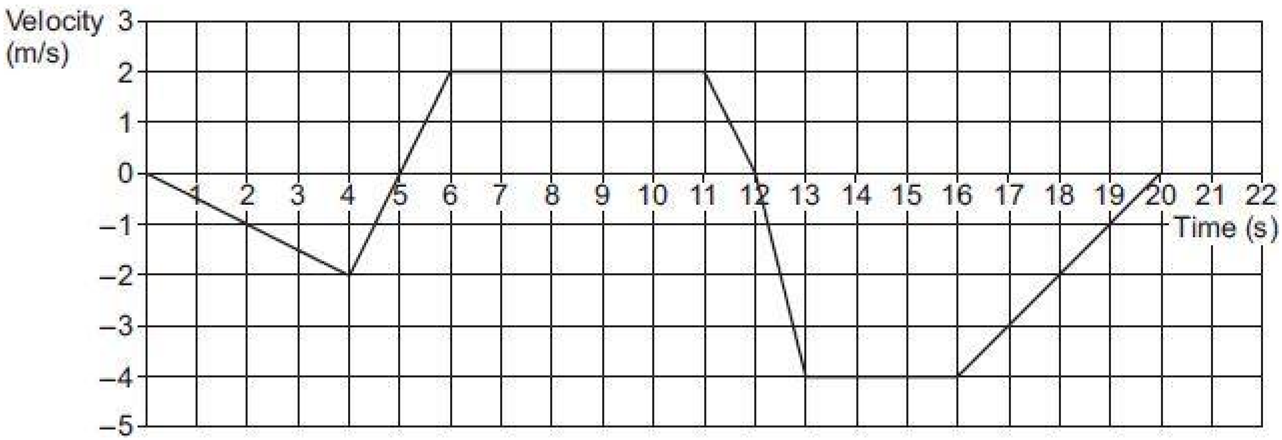
(b) Find the distance of the car from *P* after 20 seconds.

(3)

(Total 6 marks)

Q2.

The graph below shows the velocity of an object moving in a straight line over a 20 second journey.



(a) Find the maximum magnitude of the acceleration of the object.

(1)

(b) The object is at its starting position at times 0, t_1 and t_2 seconds.

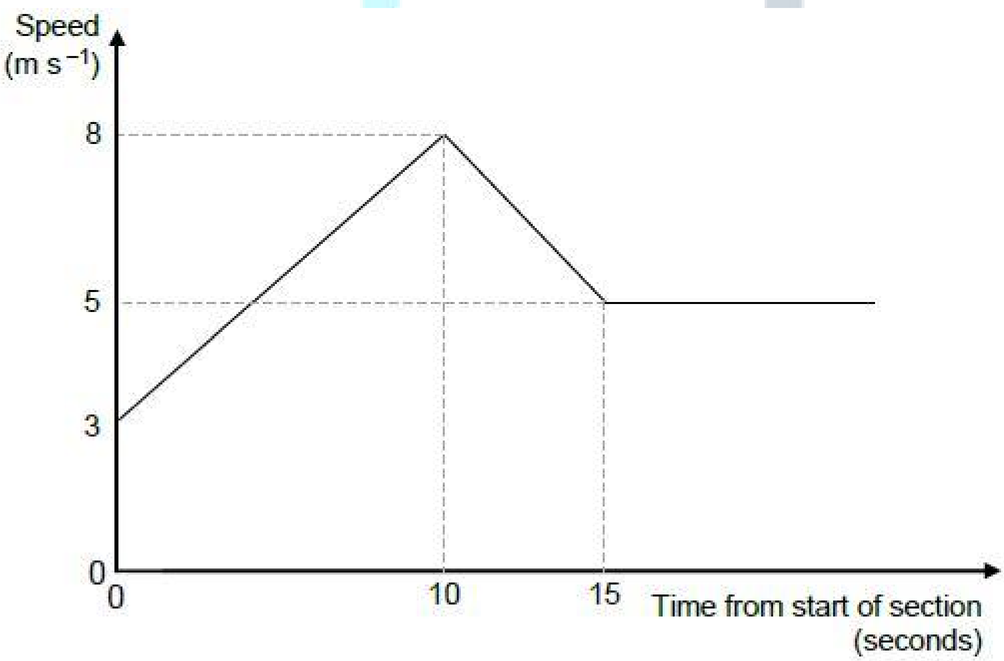
Find t_1 and t_2

(4)

(Total 5 marks)

Q3.

The graph shows how the speed of a cyclist varies during a timed section of length 120 metres along a straight track.



(a) Find the acceleration of the cyclist during the first 10 seconds.

(1)

(b) After the first 15 seconds, the cyclist travels at a constant speed of 5 m s^{-1} for a further T seconds to complete the 120-metre section.

Calculate the value of T.

(4)
(Total 5 marks)

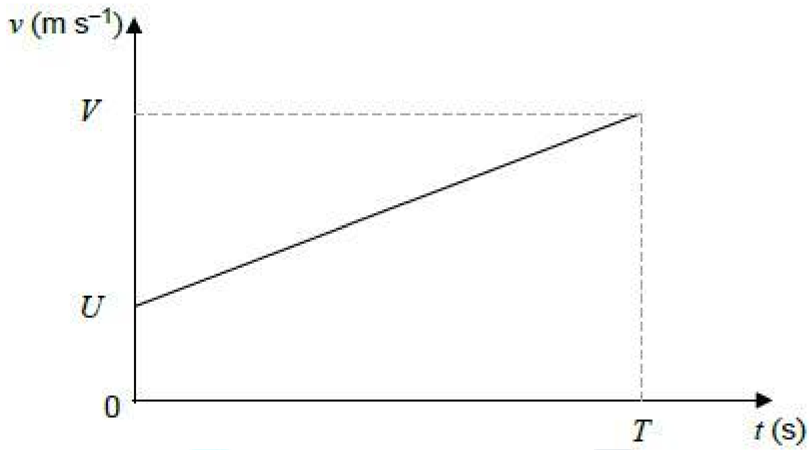
Q4.

A particle moves on a straight line with a constant acceleration, a m s $^{-2}$.

The initial velocity of the particle is U m s $^{-1}$

After T seconds the particle has velocity V m s $^{-1}$

This information is shown on the velocity-time graph.



The displacement, S metres, of the particle from its initial position at time T seconds is given by the formula

$$S = \frac{1}{2}(U + V)T$$

(a) By considering the gradient of the graph, or otherwise, write down a formula for a in terms of U, V and T.

(1)

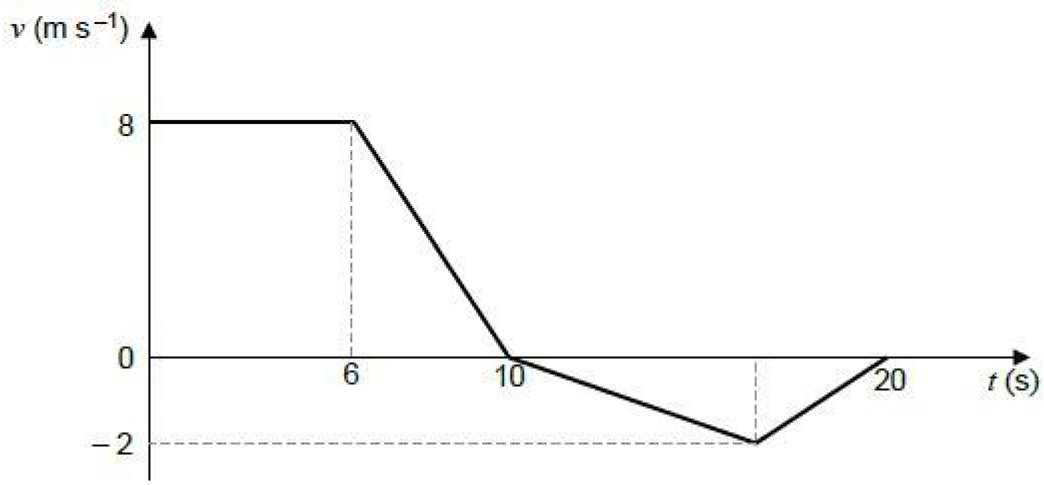
(b) Hence show that $V^2 = U^2 + 2aS$.

(3)
(Total 4 marks)

Q5.

The graph below models the velocity of a small train as it moves on a straight track for 20 seconds.

The front of the train is at the point A when $t = 0$
The mass of the train is 800kg.



(a) Find the total distance travelled in the 20 seconds.

(3)

(b) Find the distance of the front of the train from the point *A* at the end of the 20 seconds.

(1)

(c) Find the maximum magnitude of the resultant force acting on the train.

(2)

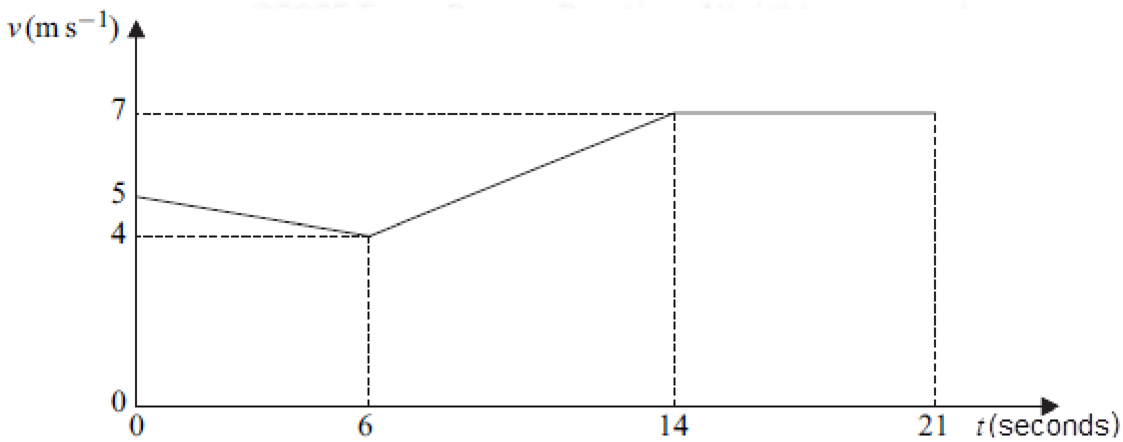
(d) Explain why, in reality, the graph may not be an accurate model of the motion of the train.

(1)

(Total 7 marks)

Q6.

The graph shows how the speed of a cyclist, Hannah, varies as she travels for 21 seconds along a straight horizontal road.



(a) Find the distance travelled by Hannah in the 21 seconds.

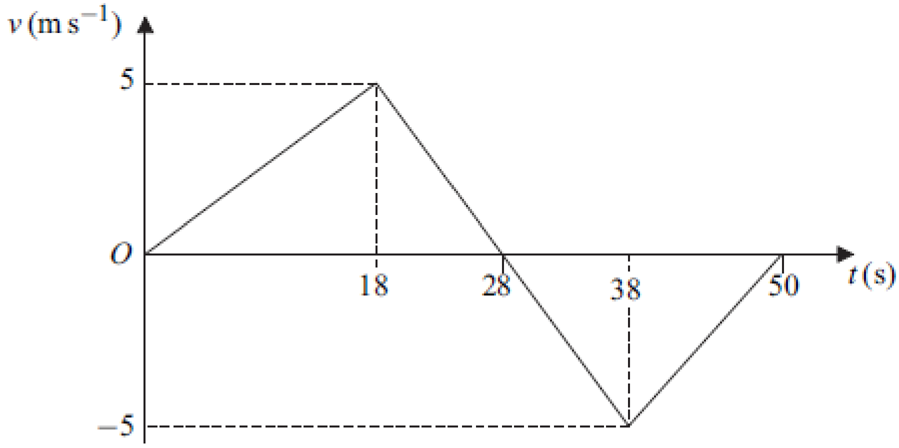
(4)

(b) Find Hannah's average speed during the 21 seconds.

(2)
(Total 6 marks)

Q7.

The diagram shows a velocity−time graph for a train as it moves on a straight horizontal track for 50 seconds.



(a) Find the distance that the train moves in the first 28 seconds.

(2)

(b) Calculate the total distance moved by the train during the 50 seconds.

(3)

(c) Hence calculate the average speed of the train.

(2)

(d) Find the displacement of the train from its initial position when it has been moving for 50 seconds.

(1)

(e) Hence calculate the average velocity of the train.

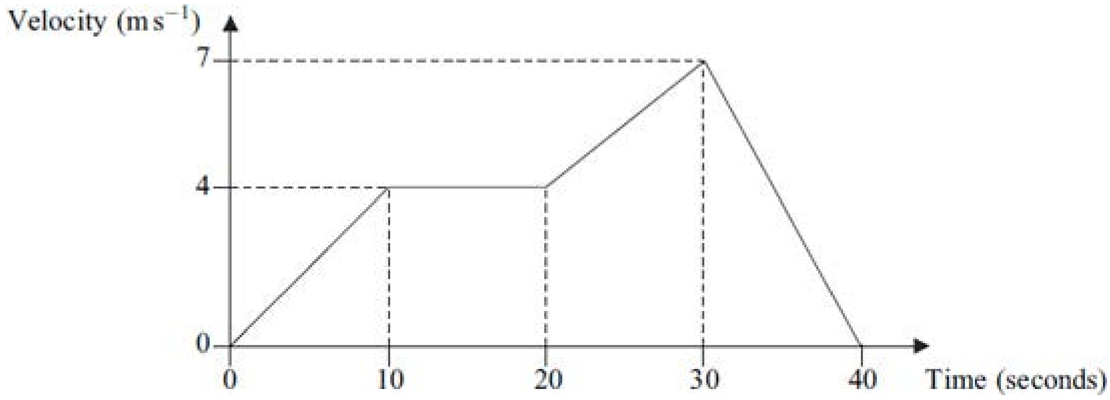
(2)

(f) Find the acceleration of the train in the first 18 seconds of its motion.

(1)
(Total 11 marks)

Q8.

The graph shows how the velocity of a train varies as it moves along a straight railway line.



(a) Find the total distance travelled by the train.

(4)

(b) Find the average speed of the train.

(2)

(c) Find the acceleration of the train during the first 10 seconds of its motion.

(2)

(d) The mass of the train is 200 tonnes. Find the magnitude of the resultant force acting on the train during the first 10 seconds of its motion.

(2)

(Total 10 marks)

Q9.

A pair of cameras records the time that it takes a car on a motorway to travel a distance of 2000 metres. A car passes the first camera whilst travelling at 32 m s^{-1}.
The car continues at this speed for 12.5 seconds and then decelerates uniformly until it passes the second camera when its speed has decreased to 18 m s^{-1}.

(a) Calculate the distance travelled by the car in the first 12.5 seconds.

(1)

(b) Find the time for which the car is decelerating.

(3)

(c) Sketch a speed–time graph for the car on this 2000-metre stretch of motorway.

(3)

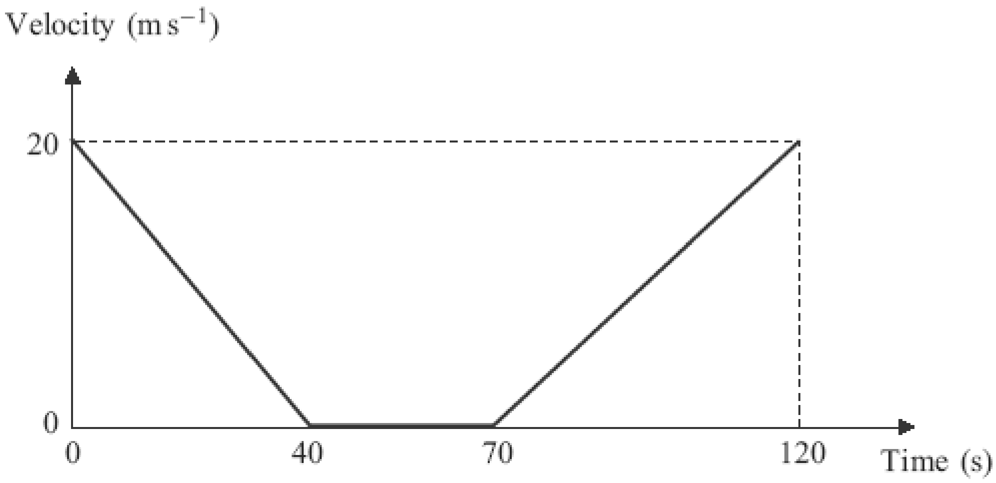
(d) Find the average speed of the car on this 2000-metre stretch of motorway.

(2)

(Total 9 marks)

Q10.

A bus slows down as it approaches a bus stop. It stops at the bus stop and remains at rest for a short time as the passengers get on. It then accelerates away from the bus stop.
The graph shows how the velocity of the bus varies.



Assume that the bus travels in a straight line during the motion described by the graph.

(a) State the length of time for which the bus is at rest.

(1)

(b) Find the distance travelled by the bus in the first 40 seconds.

(2)

(c) Find the total distance travelled by the bus in the 120-second period.

(2)

(d) Find the average speed of the bus in the 120-second period.

(2)

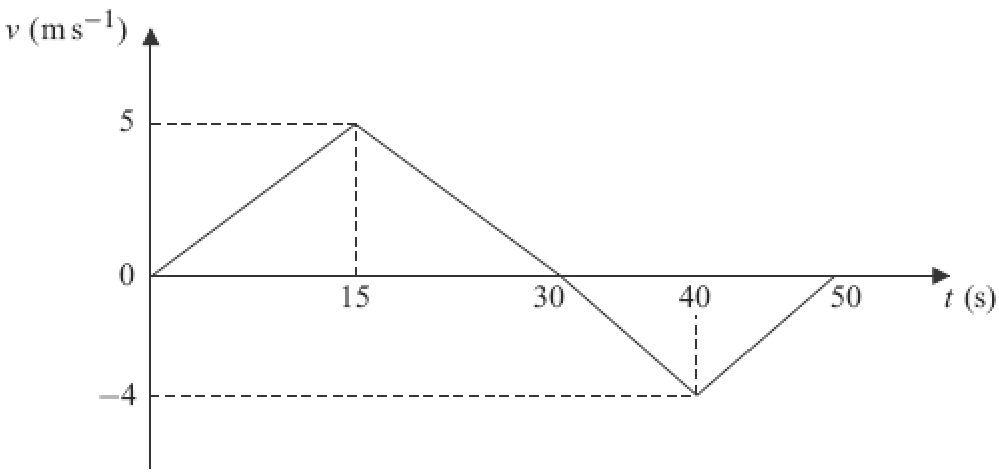
(e) If the bus had not stopped but had travelled at a constant 20 m s^{-1} for the 120-second period, how much further would it have travelled?

(2)

(Total 9 marks)

Q11.

The graph shows how the velocity of a particle varies during a 50-second period as it moves forwards and then backwards on a straight line.



(a) State the times at which the velocity of the particle is zero.

(2)

(b) Show that the particle travels a distance of 75 metres during the first 30 seconds of its motion.

(2)

(c) Find the total distance travelled by the particle during the 50 seconds.

(4)

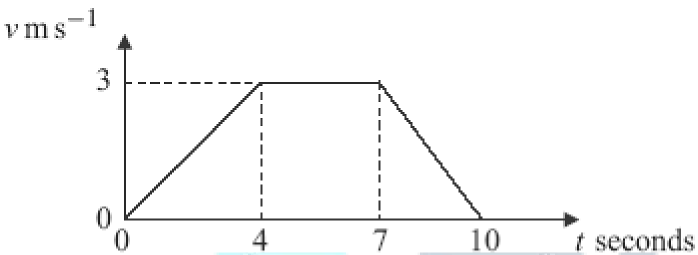
(d) Find the distance of the particle from its initial position at the end of the 50-second period.

(2)

(Total 10 marks)

Q12.

The diagram shows a velocity–time graph for a lift.



(a) Find the distance travelled by the lift.

(3)

(b) Find the acceleration of the lift during the first 4 seconds of the motion.

(1)

(c) The lift is raised by a single vertical cable. The mass of the lift is 400 kg. Find the tension in the cable during the first 4 seconds of the motion.

(3)

(Total 7 marks)

Q13.

A lift rises vertically from rest with a constant acceleration.

After 4 seconds, it is moving upwards with a velocity of 2 m s^{-1}.

It then moves with a constant velocity for 5 seconds.

The lift then slows down uniformly, coming to rest after it has been moving for a total of 12 seconds.

(a) Sketch a velocity–time graph for the motion of the lift.

(4)

(b) Calculate the total distance travelled by the lift.

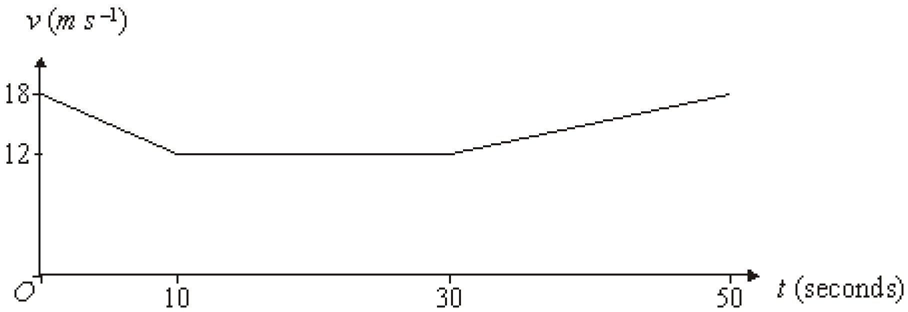
(2)

(c) The lift is raised by a single vertical cable. The mass of the lift is 300 kg. Find the maximum tension in the cable during this motion.

(4)
(Total 10 marks)

Q14.

The velocity-time graph below represents the three stages of the motion of a coach moving along a straight horizontal road. Initially the coach has velocity 18 m s^{-1}.



(a) During the first stage of the motion, the coach decelerates at a constant rate of a m s $^{-2}$ for 10 seconds until it reaches a velocity of 12 m s $^{-1}$.

(i) Find the value of a.

(2)

(ii) Find the distance that the coach travels during the 10 seconds.

(2)

(b) During the second stage of the motion, the coach travels for 20 seconds with constant velocity 12 m s^{-1}. Find the distance that the coach travels during these 20 seconds.

(1)

(c) During the third stage of the motion, the coach travels with constant acceleration, reaching a velocity of 18 m s^{-1} after a further 20 seconds.

Find the average speed of the coach during the 50 seconds of the motion.

(4)
(Total 9 marks)

Q15.

A van moves from rest on a straight horizontal road.

(a) In a simple model, the first 30 seconds of the motion are represented by three separate stages, each lasting 10 seconds and each with a constant acceleration.

During the first stage, the van accelerates from rest to a velocity of 4 m s $^{-1}$.

During the second stage, the van accelerates from 4 m s^{-1} to 12 m s^{-1}.

During the third stage, the van accelerates from 12 m s^{-1} to 16 m s^{-1}.

(i) Sketch a velocity–time graph to represent the motion of the van during the first 30 seconds of its motion.

(3)

(ii) Find the total distance that the van travels during the 30 seconds.

(4)

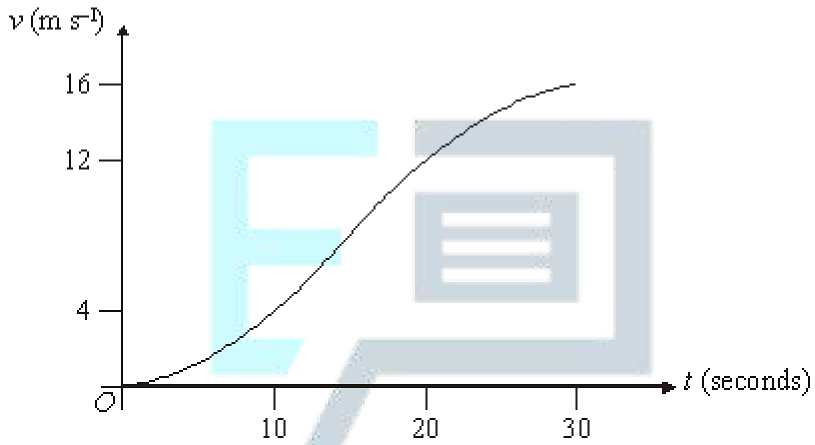
(iii) Find the average speed of the van during the 30 seconds.

(2)

(iv) Find the greatest acceleration of the van during the 30 seconds.

(2)

(b) In another model of the 30 seconds of the motion, the acceleration of the van is assumed to vary during the first and third stages of the motion, but to be constant during the second stage, as shown in the velocity–time graph below.



The velocity of the van takes the same values at the beginning and the end of each stage of the motion as in part (a).

(i) State, with a reason, whether the distance travelled by the van during the first 10 seconds of the motion in **this** model is greater or less than the distance travelled during the same time interval in the model in part (a).

(2)

(ii) Give one reason why **this** model represents the motion of the van more realistically than the model in part (a).

(1)

(Total 14 marks)

Q16.

A car travels along a straight horizontal road. The motion of the car can be modelled as three separate stages.

During the first stage, the car accelerates uniformly from rest to a velocity of 10 m s^{-1} in 6 seconds.

During the second stage, the car travels with a constant velocity of 10 m s^{-1} for a further 4 seconds.

During the third stage of the motion, the car travels with a uniform retardation of magnitude 0.8 m s^{-2} until it comes to rest.

(a) Show that the time taken for the **third** stage of the motion is 12.5 seconds.

(2)

(b) Sketch a velocity–time graph for the car during the three stages of the motion.

(4)

(c) Find the total distance travelled by the car during the motion.

(3)

(Total 9 marks)

Q17.

A car travels along a straight horizontal road. The motion of the car can be modelled as three separate stages.

During the first stage, the car accelerates uniformly from rest to a velocity of 10 m s^{-1} in 6 seconds.

During the second stage, the car travels with a constant velocity of 10 m s^{-1} for a further 4 seconds.

During the third stage of the motion, the car travels with a uniform retardation of magnitude 0.8 ms^{-2} until it comes to rest.

(a) Show that the time taken for the **third** stage of the motion is 12.5 seconds.

(2)

(b) Sketch a velocity–time graph for the car during the three stages of the motion.

(4)

(c) Find the total distance travelled by the car during the motion.

(3)

(d) State one criticism of the model of the motion.

(1)

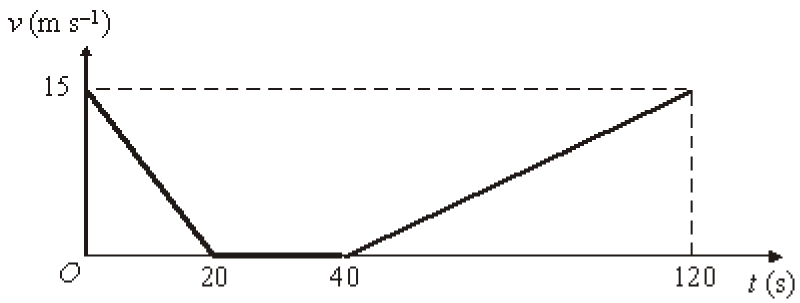
(Total 10 marks)

Q18.

A train travels along a straight horizontal track between two points A and B.

Initially the train is at A and moving at 15 m s^{-1}. Due to a problem, the train has to slow down and stop. At time t = 40 seconds it begins to move again. At time t = 120 seconds the train is at B and moving at 15 m s^{-1} again.

The graph below shows how the velocity of the train varies as it moves from A to B.



(a) Use the graph to find the total distance between the points A and B.

(4)

(b) The train should have travelled between A and B at a constant velocity of 15 m s^{-1}.

(i) Calculate the time that the train would take to travel between A and B at a speed of 15 m s^{-1}.

(1)

(ii) Calculate the time by which the train was delayed.

(1)

(c) The train has mass 500 tonnes. Find the resultant force acting on the train when $40 < t < 120$.

(4)

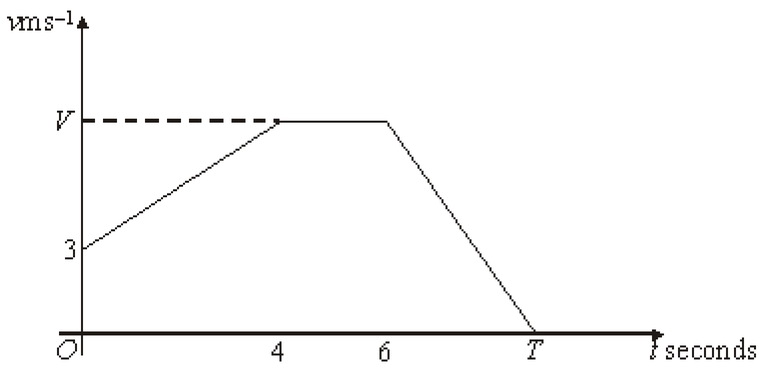
(Total 10 marks)

Q19.

The velocity–time graph below models the motion of an athlete, Paula, during part of a training exercise.

When t = 0 she has velocity 3 m s^{-1}. She accelerates at a constant rate until t = 4 and her velocity is V m s^{-1}.

She runs with velocity V m s^{-1} for 2 seconds and then decelerates at a constant rate of f m s^{-2} until she comes to rest when t = T.



(a) Between t = 0 and t = 4, Paula runs a distance of 20 metres. Show that V = 7.

(2)

(b) The total distance Paula runs between $t = 4$ and $t = T$ is 35 metres.

(i) Find the value of T.

(4)

(ii) Find the value of f.

(2)

(c) Find Paula's average speed between $t = 0$ and $t = T$.

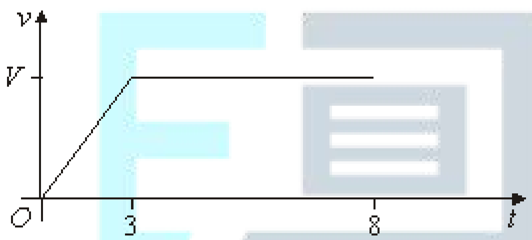
(2)
(Total 10 marks)

Q20.

The velocity-time graph below models the first 8 seconds of a journey of a bus.

During the first 3 seconds of its journey, the bus travels from rest with constant acceleration until it reaches a speed of V m s^{-1}.

During the next 5 seconds of its journey, the bus travels with constant speed V m s^{-1}.



(a) (i) During the first 3 seconds the bus accelerates at 0.8 m s^{-2}.

Show that $V = 2.4$.

(2)

(ii) Find the total distance the bus travels during the first 8 seconds of its journey.

(3)

(b) After the first 8 seconds of its journey, the bus decelerates at a constant rate until it stops. During this stage of the journey the bus travels a further 4.8 metres.

Find the average speed of the bus during the whole journey.

(4)
(Total 9 marks)

Q21.

Two particles, A and B, are connected by a light inextensible string which passes over a smooth, fixed peg, as shown in the diagram.

The particle A, of mass 0.6 kg, is in contact with a smooth horizontal surface, and the particle B, of mass 0.1 kg, hangs freely above the ground. The system is released from rest with the string taut and A moves towards the peg.

It can be assumed that, during the subsequent motion, A does **not** reach the peg.

(a) While the particles move freely, the string is taut.

(i) Show that the acceleration of the particles is 1.4 m s^{-2}.

(5)

(ii) Find the tension in the string.

(1)

(iii) Find the magnitude of the resultant force on the peg due to the tension in the string.

(3)

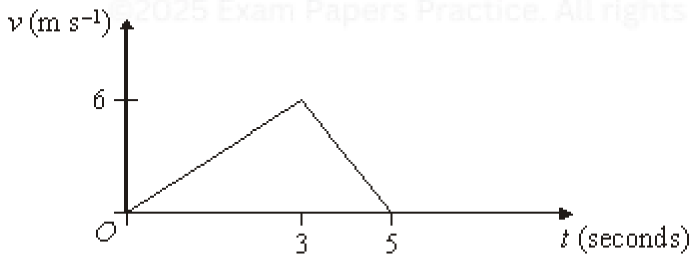
(b) After q seconds of the motion, the particle B hits the ground and remains there. The string connecting A and B slackens and A continues to move towards the peg. Sketch a velocity–time graph to show the two stages of the motion of A after being released from rest.

(3)

(Total 12 marks)

Q22.

The velocity–time graph below models Tom's motion as he runs across a playground.



(a) Find Tom's acceleration during the first 3 seconds of his motion.

(2)

(b) Find the total distance Tom runs during the 5 seconds of motion.

(3)

(c) State one criticism of this model of Tom's motion.

(1)

(Total 6 marks)

Q23.

A rabbit runs in a horizontal straight line ABC across a field.

(a) The rabbit runs from rest at A with a constant acceleration of 0.8 m s^{-2} and reaches B after 3 seconds.

Find its speed at B.

(2)

(b) The rabbit then runs from B with constant speed and reaches C after a further 4 seconds.

Sketch a velocity-time graph of the motion of the rabbit as it runs from A to C.

(3)

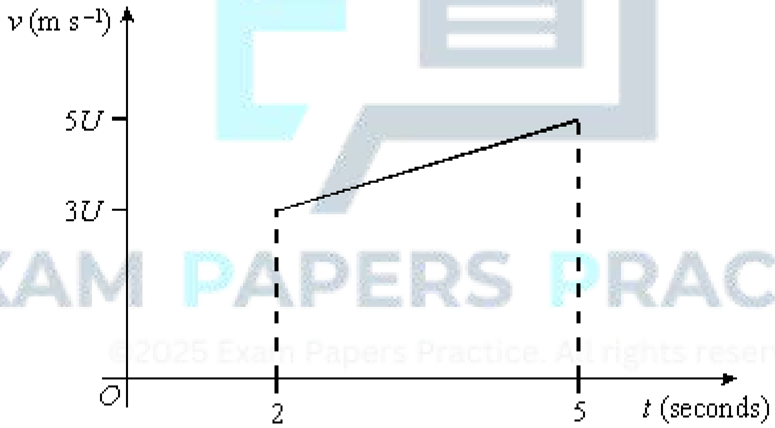
(c) Find the average speed of the rabbit as it runs from A to C.

(5)

(Total 10 marks)

Q24.

The velocity–time graph shows the motion of a particle P moving with constant acceleration.



At times t = 2 and t = 5, P has velocities $3U$ and $5U$ respectively.

(a) (i) Find, in terms of U, the acceleration of P.

(2)

(ii) Find, in terms of U, the distance travelled by P between the times t = 2 and t = 5.

(2)

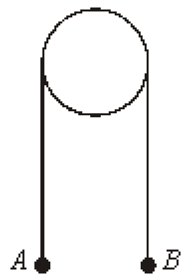
(b) When t = 5, the motion of P changes and subsequently P moves with a constant retardation. The particle P comes to rest after travelling a **further** 20 metres in the next 4 seconds.

Find the value of U.

(3)
(Total 7 marks)

Q25.

Two particles, A and B, are connected by a light inextensible string which passes over a smooth fixed peg, as shown in the diagram.



The particles A and B have masses 0.21 kg and 0.14 kg respectively. The system is released from rest. It may be assumed that, during the motion, neither particle reaches the peg.

(a) Show that the magnitude of the acceleration of the particles during the subsequent motion is 1.96 m s^{-2}.

(5)

(b) Find the speed of the particles after 2 seconds of the motion.

(2)

(c) After 2 seconds of the motion, the string breaks and the particles move independently under the influence of gravity alone. Find the time between the string breaking and the particle B coming to instantaneous rest.

(2)

(d) Sketch a velocity-time graph to show the motion of B between being released from rest and subsequently coming instantaneously to rest.

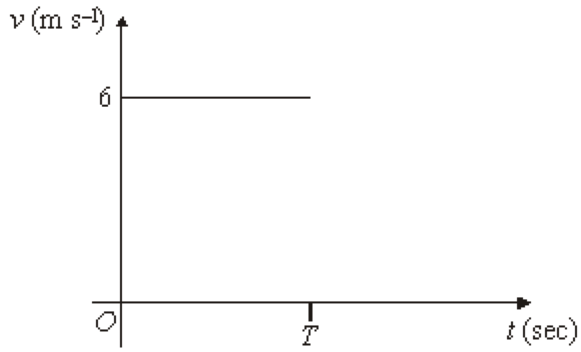
(3)
(Total 12 marks)

Q26.

Fiona is cycling along a straight horizontal road when she spots her friend, Marie, ahead of her. Marie is jogging along the same road and in the same direction.

Fiona is cycling with constant speed 6 m s^{-1} and Marie is jogging with constant speed 3 m s^{-1}.

(a) The velocity-time graph below shows Fiona's motion from the moment when she spots Marie, when t = 0, until she reaches a bridge, when t = T.



Marie takes 2 seconds longer than Fiona to reach the bridge.

Copy the graph and draw another line to show Marie's motion from t = 0 until she reaches the bridge. Label the axes appropriately.

(2)

(b) When t = 0, Marie is x metres from the bridge and Fiona is 20 metres behind her.

(i) Write down, in terms of x, the distance travelled by Fiona.

(1)

(ii) Write down an equation in terms of x and T for Fiona's motion.

(1)

(iii) Write down an equation in terms of x and T for Marie's motion.

(2)

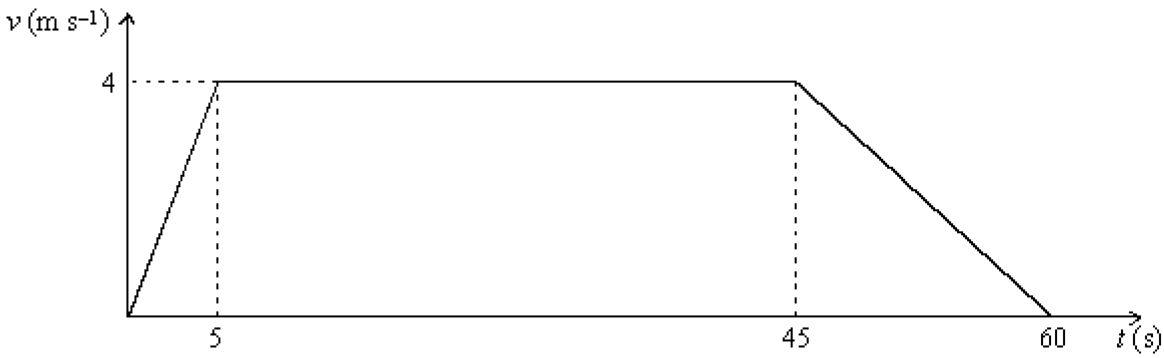
(iv) Hence find the values of x and T.

(4)

(Total 10 marks)

Q27.

The graph shows how the velocity, v m s^{-1}, of a cyclist varies with time, t seconds, as she moves along a straight horizontal road.



(a) Calculate the total distance travelled by the cyclist.

(3)

(b) Find the acceleration of the cyclist during the first 5 seconds of the motion.

(2)

(c) As the cyclist moves, a force of magnitude P newtons acts on the cyclist in the direction of motion. A constant horizontal resistance force of magnitude 50 newtons also acts. Model the cyclist and her bicycle as a particle of mass 65 kg.

(i) Draw a diagram to show **all** the forces acting on the particle.

(1)

(ii) Find the value of P during the first 5 seconds of the motion.

(3)

(Total 9 marks)